

Research Report: How Traffic Affects Uber Pricing

Location Taken: Gachibowli, Hyderabad

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Introduction:

I have taken Gachibowli as my reference area because I stay near Yamjal and I often travel to Gachibowli. Everyone knows Gachibowli is the main IT hub in Hyderabad with DLF buildings, Wipro circle, Financial District and colleges like IIIT and ISB. So traffic here is very heavy especially in morning and evening time. In this report I am explaining how traffic directly changes Uber and other cab fares. Traffic not only makes us late but also makes the ride costly for passengers.

Main Reasons How Traffic Increases Fare

1. Surge Pricing - Main Reason:

I observed in Uber app when there is heavy traffic near Gachibowli circle or ORR junction, the price suddenly shows 1.5x or 2.0x surge. The reason is simple, as per the articles we were given, Uber uses GPS tracking and data science to check how many cars are stuck and how many people are requesting rides at same time.

When my dataset shows high vehicles (90 to 110 vehicles at 6 PM at Gachibowli junctions), there are less cabs free because all cabs are stuck in same traffic. But demand is high because office people are leaving at same time. So Uber automatically increases price. For example, normal fare from Gachibowli to Hitech City is around Rs. 150 but in 7 PM traffic it shows Rs. 280 to Rs. 320.

I read in moldstud article that prediction uses sensors, IoT devices and road cameras to collect traffic intensity. Uber also does same.

2. Time Charge Becomes More:

Uber fare is not only distance, it also charges for time. The formula is Base Fare + km charge + time charge. In Gachibowli to Airport road, normally it takes 20 to 25 minutes, but in peak traffic it takes 50 minutes. So time charge almost doubles. Even if distance is same, we pay extra Rs. 40 to 50 just for time.

As per avutec website, they explained that traffic data collection using telecom networks helps to calculate this travel time delay and it is added in fare.

3. Long Route Due to Traffic Jam:

Sometimes when there is Ganesh festival or any big event near Gachibowli, police closes the main road. Then Uber driver takes long route via Kokapet or Financial District inner roads. So 5 km ride becomes 7 km. So distance fare also increases because of traffic.

What Happens to Passengers and Drivers

For passengers like us in Gachibowli, we have to pay 40 to 60% more in peak hours which is 9 to 11 AM and 6 to 9 PM. Also cab takes long time to come, ETA shows 10 to 12 mins instead of 2 to 3 mins.

For drivers, they get more money per trip during surge but they can do less trips. In normal time they can do 3 trips in one hour but in Gachibowli traffic they do only 1 or 2 trips. Also petrol cost is more due to stop and go traffic.

Overall Business Effect on Uber

Uber's main income comes from this surge pricing only. In big cities like Hyderabad almost 40% revenue is from peak hour surge only. If Uber can predict traffic before only, like using our project traffic forecast data, they can send more cabs to Gachibowli before 6 PM and control price and ETA. Also customer satisfaction will increase if fare is shown correctly before booking.

This is why our project to forecast hourly traffic volume at different junctions is useful for Uber business, not just for traffic police.

Conclusion:

So traffic is directly connected to Uber fare. More traffic means more fare. By forecasting hourly traffic at junctions we can help Uber to predict surge in advance and help passengers to plan better. For Gachibowli area this is very important because daily thousands of IT employees depend on cabs.